

**Birla Institute of Technology & Science, Pilani**  
**Work Integrated Learning Programmes Division**  
**First / Second Semester 2023-2024**

**Mid-Semester Test**  
**(EC-2 Makeup)**

Course No. : DSECLZG529/AIMLCZG529  
Course Title : Data Management for Machine Learning  
Nature of Exam : Closed Book  
Weightage : 30%  
Duration : 2 Hours  
Date of Exam : 28/7/2024 (AN)

|                  |     |
|------------------|-----|
| No. of Pages     | = 2 |
| No. of Questions | = 4 |

**Note to Students:**

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q.1) Assume you are developing a Fitness and Wellness app that allows users to track their workouts, monitor their diet, set fitness goals, and receive personalized health recommendations. The app aims to fulfill all fitness and wellness needs, offering features such as exercise logging, meal tracking, water intake monitoring, sleep tracking, and mindfulness exercises. Users can also connect with personal trainers and nutritionists, join fitness challenges, and access a library of workout videos and healthy recipes. The app should include listings from certified trainers and dietitians, user-generated reviews, and no-commission options. When users are online, the app should provide recommendations for relevant fitness programs and display targeted advertisements to enhance user experience and increase ad conversion rates. **[1.5 + 2 + 3 + 2 + 1 + 1 + 2 + 2.5 = 15]**

a) What type of data processing is suitable for the above requirement? Why? [1.5 M]

b) Discuss the important data modalities that are suitable for such applications. [2 M]

c) Who are the primary users of this system and what type of user interfaces are suitable for them to interact with this application? [3 M]

d) Which data format will be suitable to exchange the data between users of the application and the recommendation engine? Provide a snippet of the data item exchanged. [2 M]

e) Assume a personal trainer is promoting their fitness programs on this application and needs a periodic report on user engagement with their workout videos, number of users following their plans, feedback received, and overall user progress. What type of data abstraction will be required to generate such a report? Why? [1 M]

f) Imagine that now you have to implement a feature where immediate fitness program recommendations need to be shown to the users when they are logging their activities. What significant change will be required in the data flow mentioned in (f) to match the new requirement? [1 M]

g) Identify any two approaches through which this prepared report can be served to the personal trainer or their representative. [2 M]

h) Discuss briefly the data flow (and the stages involved) that will be required to meet this requirement along with a pictorial representation. [2.5 M]

Q.2a) A healthcare organization needs to address the following challenges with Big Data Analytics:

- Unable to relate raw data collected from medical devices, patient monitoring systems, or wearable health trackers with patient records and administrative data managed in enterprise systems.
- Each end-to-end ETL process consumes 10 hours. This process should be completed within 30 minutes.
- Struggling with a Python-based ETL tool to manage and extract value from the growing volume and variety of patient data and need to unify information across federated sources. Each disparate PostgreSQL server records around 1 million updates, a total of 30 million updates daily.
- Some data warehousing queries on structured data are available only in SQL and there is no expertise available for transforming these SQL queries into Hadoop ecosystem tools.

Recommend suitable solutions for the above challenges by making use of Hadoop Ecosystem products/tools. Give justifications for your recommendations.

**[1+1+1+1 = 4 marks]**

Q.2b) What Hadoop ecosystem tool should be used for efficient real-time stream processing of big data and why?

**[3 marks]**

Q.3) Observe the following table of the data of the Government Engineering College, Palampur Students. The applicable Fee for the students is Rs. 70,000 per annum. The application is for predicting the fee of the students. Is there any data leakage present? If yes , explain in detail about the data leakage.

**[4 marks]**

| Roll No. | Name          | Category | ..... | Location | Total Applicable fee (Rs.)<br>(Target Column) |
|----------|---------------|----------|-------|----------|-----------------------------------------------|
| 1111     | Uttpal Dutta  | Open     | ..... | Pune     | 70,000                                        |
| 1112     | Samir Shaikh  | OBC      | ..... | Kolkata  | 35,000                                        |
| 1113     | Amitabh Kumar | OPEN     | ..... | Pune     | 70,000                                        |
| .....    | .....         | .....    | ..... | .....    | .....                                         |
| 7657     | Anjuja Tiwari | OBC      | ..... | Indore   | 35,000                                        |
| 7658     | Ram Varma     | OPEN     | ..... | Indore   | 70,000                                        |
| 7659     | Sunny Deol    | OBC      | ..... | Jaipur   | 35,000                                        |

Q.4) When you will select the document model for your database. Explain it with suitable example ?

**[4 marks]**

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